Topic: Table of transforms

Question: Use a table of Laplace transforms to transform the function.

$$f(t) = 5t - 6t^2$$

Answer choices:

A $$F(s) = \frac{5}{s} - \frac{6}{s^2}$$

B $$F(s) = \frac{5}{s^2} - \frac{6}{s^3}$$

C $$F(s) = \frac{5}{s} - \frac{12}{s^2}$$

D $$F(s) = \frac{5}{s^2} - \frac{12}{s^3}$$

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Solution: D

From the table of Laplace transforms, we know

$$\mathscr{L}(t^n) = \frac{n!}{s^{n+1}}$$

With $n = 1$ and $n = 2$, we find

$$\mathscr{L}(t) = \frac{1}{s^2}$$

$$\mathscr{L}(t^2) = \frac{2}{s^3}$$

Then the Laplace transform of $f(t) = 5t - 6t^2$ is

$$F(s) = 5\left(\frac{1}{s^2}\right) - 6\left(\frac{2}{s^3}\right)$$

$$F(s) = \frac{5}{s^2} - \frac{12}{s^3}$$

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Topic: Table of transforms

Question: Use a table of Laplace transforms to transform the function.

$$f(t) = e^{3t} + 4$$

Answer choices:

A $$F(s) = \frac{1}{s+3} - \frac{1}{4s}$$

B $$F(s) = \frac{1}{s-3} + \frac{1}{4s}$$

C $$F(s) = \frac{1}{s-3} + \frac{4}{s}$$

D $$F(s) = \frac{1}{s+3} - \frac{1}{s}$$

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Solution: C

From the table of Laplace transforms, we know

$$\mathscr{L}(e^{at}) = \frac{1}{s-a}$$

$$\mathscr{L}(1) = \frac{1}{s}$$

With $a = 3$, we find

$$\mathscr{L}(e^{3t}) = \frac{1}{s-3}$$

Then the Laplace transform of $f(t) = e^{3t} + 4$ is

$$F(s) = \frac{1}{s-3} + 4\left(\frac{1}{s}\right)$$

$$F(s) = \frac{1}{s-3} + \frac{4}{s}$$

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Topic: Table of transforms

Question: Use a table of Laplace transforms to transform the function.

$$f(t) = 5 + 3t + t^2$$

Answer choices:

A $F(s) = 5 + \frac{3}{s} + \frac{2}{s^2}$

B $F(s) = \frac{5}{s} + \frac{3}{s^2} + \frac{2}{s^3}$

C $F(s) = \frac{5}{s} + \frac{3}{s^2} + \frac{1}{s^3}$

D $F(s) = 5 + \frac{3}{s} + \frac{1}{s^2}$

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Solution: B

From the table of Laplace transforms, we know

$$\mathscr{L}(t^n) = \frac{n!}{s^{n+1}}$$

$$\mathscr{L}(1) = \frac{1}{s}$$

With $n = 1$ and $n = 2$, we find

$$\mathscr{L}(t) = \frac{1}{s^2}$$

$$\mathscr{L}(t^2) = \frac{2}{s^3}$$

Then the Laplace transform of $f(t) = 5 + 3t + t^2$ is

$$F(s) = 5\left(\frac{1}{s}\right) + 3\left(\frac{1}{s^2}\right) + \frac{2}{s^3}$$

$$F(s) = \frac{5}{s} + \frac{3}{s^2} + \frac{2}{s^3}$$

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